CASE STUDY PRESENTATION • CLOUD COMPUTING ESSENTIALS (CCE)

LabTrack

Hardware Lifecycle Governance & Containerized Laboratory Automation for SIT Pune

Autonomous Microservices Architecture for Symbiosis Institute of Technology Engineering Labs

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COURSE EVALUATION • CLOUD COMPUTING ESSENTIALS

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Course: Cloud Computing Essentials (CCE • 2 Credits, 30 Marks)

PROBLEM IMPACT & EVIDENCE

Academic hardware inefficiency is already impacting SIT Pune

Real-world operational evidence across Symbiosis Institute of Technology engineering laboratories demonstrates severe fiscal and operational failure.

R&A LABS 2026 — CAPSTONE INVENTORY AUDIT

Department Reports 18 Development Boards & Sensors Untracked in Capstone Stocktaking

Physical lab stocktaking in Robotics & Automation labs revealed unreturned Jetson Nanos, LiDARs, and sensor modules across graduating capstone batches. Absence of authenticated identity bindings makes retrospective recovery impossible.

Audited from Capstone Project Inventory

STUDENT CAPSTONE GRIEVANCE — SENIOR BATCH LOG

Senior Capstone Project Blocked for Weeks Awaiting Basic Ultrasonic Sensors

Recorded grievance from an SIT senior whose capstone milestone was blocked because prior student teams hoarded basic ultrasonic sensors and development boards in personal lockers with zero automated return enforcement.

Senior Capstone Milestone Stagnation

CODEX TECHNICAL CLUB — SYMBITECH 2026

CodeX Flagship Event (SymbiTech) Delayed Due to Missing Hardware Components

During SIT's flagship technical symposium SymbiTech, the CodeX technical club was forced to execute their flagship event late because required sensor modules and boards could not be located in central storage due to lack of visibility, missing checkout registers, and untracked returns.

SymbiTech Flagship Execution Delay

Core Takeaway: Traditional paper logbooks and un-synced spreadsheets failed SIT Pune's engineering labs, resulting in capstone milestone delays, unlocatable hardware during SymbiTech, and untracked lockers.

Problem definition: SIT Pune hardware lifecycle governance

SYSTEM DOMAIN & OPERATIONAL AIM (SDD V0.1)

Replaces fragmented email, notebook, and spreadsheet handling with project-linked availability, approval, custody, and reconciliation history specifically for SIT Pune engineering laboratories.

- **SIT Laboratory Scope:** Robotics & Automation and AI & Machine Learning department laboratories.
- **Non-Disposal Rule:** Consumables and components are returnable and reusable, not disposed on checkout.
- **Eliminate Paper Register Books:** Authenticated digital custody bound to verified student PRNs and capstone project IDs.
- **Prevent Inventory Corruption:** Enforce atomic SQL decrements with row-level locks ($Q_{available} \geq Q_{requested}$).
- **Multi-Tier Academic Approval:** Consumables route to Mentor; high-value assets strictly require Mentor + Lab In-Charge + HoD sign-off.

THE HARDWARE CUSTODY ICEBERG

Visible Surface: Simple Check-In / Check-Out

What naive commercial IT asset tools see: static barcode scans and binary in/out switches.

Submerged Realities Solved by LabTrack:

- **14-Stage SDD Lifecycle:** Project → Requirement → Availability → Request → Faculty Review → Class Routing → HoD Sign-off → Dispatch → Tracking → Reminder → Return → Reconciliation → Stock Update → History.
- **Capstone Role Hierarchy:** Multi-tier approval routing based on academic project lead identity.
- **Concurrency Contention:** Database row locks preventing 2 students from claiming the final sensor.
- **Return Station Math:** Strict condition segregation: $Q_{good} + Q_{defective} = Q_{issued}$.

System Aim: Unify SIT Pune hardware governance into an auditable, cloud-ready software architecture that guarantees zero inventory drift.

Core architecture: decoupled microservices & academic governance

MICROSERVICES ARCHITECTURAL FOUNDATIONS

1. Microservices Architecture:
Decoupled independent service containers: FastAPI REST service, asynchronous background worker container for SMTP/events, and one-shot Alembic migration init container.

2. SQL ACID Persistence:
Relational store (22 tables) strictly owns business truth and inventory state. Row-level locks (`SELECT ... FOR UPDATE`) prevent concurrency race conditions.

3. Role-Based Security & Encryption:
Salted SHA-256 password hashing + stateless Bearer JWT session tokens with role claims + 4-tier RBAC (Student, Faculty Mentor, Lab In-Charge, HoD).

4. Code Modularity & Campus Deployability:
Modular codebase with clean separation of concerns, architected for seamless containerized deployment on SIT Pune's on-premises Linux bare metal infrastructure.

ACADEMIC LITERATURE REVIEW & THEORETICAL BASIS

NLP Semantic Triage (Loper & Bird, TextBlob):
Applies lexical sentiment and keyword polarity analysis to requisition justifications, classifying requests into routine vs safety-critical/urgent capstone tiers.

Academic Resource Prioritization (IEEE Models):
Formal multi-criteria priority queues balancing student academic standing, capstone milestones, and component scarcity over unvetted manual approvals.

Least Privilege RBAC in Distributed Systems (NIST SP 800-162):
Cryptographic role boundaries preventing student privilege escalation to faculty approval or custodian tiers across distributed API endpoints.

Architectural Invariant: Decoupled microservices isolate client latency from background tasks, while NLP semantic triage prioritizes critical academic requests.

REQUIREMENTS TAXONOMY

Functional & non-functional specifications (SDD v0.1)

FUNCTIONAL REQUIREMENTS (FR)

- **REQ-LT-0001 (Role-Based Access):** 4 academic personas (Student, Faculty Mentor, Lab In-Charge, HoD) with cryptographically validated permissions.
- **REQ-LT-0016 (Atomic Decrement):** Row locks prevent negative inventory under concurrent multi-student checkout requests.
- **REQ-LT-0035 (FIFO Reservation Engine):** Automatic queue advancement when items are returned or restocked.
- **REQ-LT-0042 (Physical Return Inspection):** Segregates returned parts into active stock vs repair quarantine.
- **REQ-LT-0059 (Autonomous Outbox Worker):** Dedicated background worker polling SQL outbox records for event relay.

NON-FUNCTIONAL REQUIREMENTS (NFR)

- **NFR-SEC-01 (Credential Security):** Passwords hashed with salted SHA-256; stateless Bearer JWT session tokens with role claims.
- **NFR-REL-01 (Transactional Integrity):** Zero inventory drift under concurrent peak checkout spikes through database ACID guarantees.
- **NFR-PERF-01 (Sub-50ms Response):** FastAPI ASGI server delivers sub-50ms response times for operational catalog queries.
- **NFR-ISO-01 (Container Hardening):** Non-root execution (`uid 10001:10001`) with memory and CPU boundaries.
- **NFR-AUD-01 (Immutable Auditability):** Append-only SQL audit logging tracking all state transitions.

Specification Guarantee: Formal requirements in SDD v0.1 ensure that core domain invariants remain intact across all environments.

End-to-end hardware lifecycle & novel user flow

01. REQUISITION	02. SEMANTIC TRIAGE	03. ROLE APPROVAL	04. COUNTER DISPATCH	05. RETURN STATION
Student Kit Planning	TextBlob NLP Classification	Multi-Tier Academic Chain	Barcode Custody Binding	Condition Reconciliation

01	**Student Kit Planning**	Student selects hardware for their registered Capstone project. Server validates availability in real time; unavailable items automatically prompt FIFO queue enrollment.
02	**Semantic Triage (TextBlob NLP)**	NLP evaluates justification text and component cost. High-value robotic assets are elevated into a Priority Queue visible to the Department HoD.
03	**Multi-Tier Academic Chain**	Consumables route through Faculty Mentor; high-value assets strictly require Mentor + Lab In-Charge + HoD sign-off.
04	**Counter Barcode Binding**	Lab Assistant scans physical SIT barcode (e.g. SIT-ROBOT-0042) at the counter. Atomic SQL decrement commits physical custody to the student group.
05	**Return Station Math**	Assistant inspects returned kit. 4 sensors good, 1 burnt. Math reconciles: 4 to active stock, 1 to repair quarantine. Student reputation score updated.

Novelty: Replaces informal verbal handovers with a deterministic 5-stage digital custody chain with verifiable audit history.

Cloud infrastructure & enterprise service integration

ARCHITECTURE LAYER	SERVICE SELECTED	CLOUD ROLE & RESPONSIBILITIES	ENTERPRISE CLOUD TARGET
Edge Frontend Hosting	Vercel CDN	Hosts React 19 single-page app on global edge points of presence with GitHub CI/CD automation	AWS CloudFront / Azure Front Door
Managed Compute	Azure Container Apps	Serverless container runtime for FastAPI & Worker; automatically scales to zero during vacations	AWS ECS Fargate
Relational Business Truth	Azure MySQL Flexible	Authoritative ACID store (22 tables) enforcing row-level locks, foreign keys, and audit logs	AWS RDS MySQL Multi-AZ
Document Read Model	MongoDB Atlas	Managed cloud NoSQL database serving notification feeds, audit timelines, and inspection images	AWS DocumentDB
Asynchronous Outbox	Transactional Outbox	Decouples SQL stock commits from external Gmail SMTP; models AWS SQS FIFO reliability	AWS SQS FIFO Queues
Security Perimeter	App Gateway WAF_v2	OWASP CRS 3.2 ruleset; deep inspection of payloads blocking SQLi, XSS, and parameter tampering	AWS WAF

Hybrid Cloud Strategy: Demonstrates cloud deployment readiness (Azure Container Apps, AWS Outbox, Vercel Edge) while code runs verified locally in Docker.

SYSTEM CONTAINERIZATION

Containerization: multi-service docker architecture

SERVICE IN COMPOSE.YAML	CONTAINER ROLE IN LABTRACK	WHY IT IS CONTAINERIZED	PORT & SECURITY BOUNDARY
mysql	Authoritative relational store for inventory, users, and audit logs	Guarantees identical relational engine without polluting host OS	Port 3306; persistent named volume
mongo	Document store for user notifications and event projections	Provides isolated document database for read-model caching	Port 27017; persistent volume
migrate	Runs Alembic schema migrations and applies seed catalogs	One-shot init container; prevents database race conditions on boot	Ephemeral; exits with code 0 on finish
api	FastAPI REST service running Uvicorn ASGI runtime	Packages Python 3.12 dependencies under secure non-root uid 10001	Port 8000; HEALTHCHECK enabled
worker	Asynchronous background processor polling SQL outbox records	Isolates slow background tasks from fast API response threads	Internal only; no exposed ports

Container Hardening: Non-root execution (`uid 10001:10001`), multi-stage builds reducing image size to ~180MB.

Literature on Container Virtualization: Felter et al. (IBM Research / IEEE 2015) & Burns et al. (ACM Queue 2016): Proves near bare-metal I/O throughput vs hypervisors.

DATA ARCHITECTURE

Polyglot persistence & department excel sync engine

RELATIONAL BUSINESS TRUTH (SQL)

Authoritative Single Source of Truth

- **22 Relational Tables:** Users, Projects, Assets, Allocations, Outbox, and Audit Logs.
- **Authentication Security:** Passwords hashed using SHA-256 with per-user cryptographic salts.
- **Concurrency Control:** Strict row-level locking (`SELECT ... FOR UPDATE`) guarantees zero stock over-allocation.
- **Atomic Return Math:** Single transaction commits restock and quarantine: $Q_{good} + Q_{defective} = Q_{issued}$.

DEPARTMENT EXCEL SYNC & MONGODB

Administrative Utility & Read Projections

- **Department Excel Sync (utilities/sync_frontend_mockdata.py):** Lab assistants manage hardware catalogues in spreadsheets. Python utility validates schemas, hashes sensitive data, and syncs the live database.
- **MongoDB Atlas Document Store:** 7 collections serving unstructured notification feeds, equipment inspection images, and event timelines.
- **CQRS Academic Literature (Martin Fowler):** Command Query Responsibility Segregation strictly separates relational ACID transactions from document read feeds.

> **Data Rule:** Inventory counts and approvals never mutate in MongoDB; transactional integrity is strictly owned by SQL.

Event reliability: transactional outbox & SMTP lifecycle

THE DUAL-WRITE PROBLEM & RESOLUTION

Updating inventory in SQL and synchronously invoking an external Gmail SMTP server creates an unrecoverable failure point if the network times out.

- **Atomic Local Outbox Commit:** FastAPI writes the inventory decrement AND an outbox event in the same transaction (<15ms).
- **Counter Handover Unblocked:** The student receives physical hardware immediately without waiting for mail delivery.
- **Asynchronous Worker Relay:** Background worker polls outbox records and dispatches Gmail SMTP with exponential backoff.

LITERATURE SURVEY & SYSTEM SPECIFICATIONS

AWS Prescriptive Guidance (2023):
Transactional Outbox Pattern for Guaranteed Event Reliability across distributed service boundaries without 2PC locking.

IETF RFC 5321 (2008):
Simple Mail Transfer Protocol specifications for reliable store-and-forward email relay architecture.

Comparative Literature (Snipe-IT & GLPI):
Open-source IT tools lack transactional outbox decoupling and stall under high-concurrency university checkouts.

Live Demonstration Available: Interactive end-to-end checkout & outbox event dispatch is verified on Slide 15. `http://localhost:5173`

Resilience Guarantee: Physical hardware checkout is never blocked or rolled back by third-party email server latency (AWS Prescriptive Guidance).

Intelligent allocation: FIFO, TextBlob NLP & reputation

DETERMINISTIC QUEUING

FIFO Reservation Worker

Out-of-stock items automatically enroll projects into a deterministic queue (`created_at, request_id`). When hardware is reconciled at return, the worker automatically reallocates stock to the next eligible team.

ZERO FAVORITISM • FAIR ACCESS

SEMANTIC NLP TRIAGE

TextBlob Tier Classifier

Evaluates requisition justification text and component value using NLP. High-complexity or safety-critical capstone requests are elevated into a Priority Queue visible to the Department HoD.

PRIORITY QUEUE FOR HOD

BEHAVIORAL RELIABILITY

Student Reputation Score

Computes a dynamic reliability metric based on historical on-time returns, defect frequency, and inspection condition. Responsible teams receive expedited processing; chronic defaulters require mentor co-sign.

DYNAMIC TRUST METRIC

Live Demonstration Available: View dynamic student reputation scoring and NLP triage queue live on Slide 15. `http://localhost:5173`

Algorithmic Novelty: Intelligent queuing and semantic prioritization transform static inventory management into an adaptive academic control plane.

Institutional impact & circular governance at SIT Pune

QUANTIFIED SIT PUNE INSTITUTIONAL VALUE

- **100% Paperless Auditability:** Full digital traceability across student PRNs and capstone project IDs for SIT department accreditation and review boards.
- **Zero Technical Event Stagnation:** Prevents component unavailability and execution delays for campus technical flagships like SymbiTech (CodeX Club).
- **Circular Defect Triage:** Physical return inspection separates repairable components (e.g. blown diodes) into triage quarantine instead of discarding.
- **120+ Staff Hours Saved Annually:** Eliminates manual paper register audits and end-of-semester physical stocktaking searches in student lockers.
- **Zero Recurring Cloud Hosting Bills:** Seamless deployment onto SIT Pune's existing bare metal campus server infrastructure eliminates ongoing monthly cloud hosting fees.

CIRCULAR HARDWARE GOVERNANCE CYCLE

1. Procure & SIT Barcode Tag:
Verified asset entry bound to department inventory and serial records.

2. Capstone Project Allocation:
Role-approved issuance bound to registered student team and PRN.

3. Physical Return Inspection:
Strict condition math: $Q_{good} + Q_{defective} = Q_{issued}$.

4. Restock or Repair Triage:
Circular hardware recovery: salvage diodes & sensors on repair bench.

Institutional Value: LabTrack transforms university hardware from an untracked consumable expense into a circular, auditable institutional asset.

Production Roadmap: Azure VM Dry Run to SIT Bare Metal

A phased engineering deployment pipeline transitioning verified local microservices to SIT Pune campus Linux bare metal servers.

STAGE 01 — VERIFIED RUNTIME

Local Docker Compose

5 OCI microservice containers (`api`, `worker`, `migrate`, `mysql`, `mongo`) running under non-root execution (`uid 10001:10001`). Verified across 61 automated pytest test suites covering ACID row locks, outbox worker polling, and defect triage math.

100% Local Proof of Work

STAGE 02 — STAGING DRY RUN

Azure Cloud VM Staging

Containerized validation on Azure Linux Virtual Machine. Validates public DNS resolution, reverse proxy routing, TLS termination, and live Gmail SMTP dispatch over public networks with exponential backoff and connection retry safeguards.

Network & TLS Validation

STAGE 03 — CAMPUS PRODUCTION

SIT Bare Metal Production

Direct migration to SIT Pune's on-premises Linux bare metal server infrastructure. Eliminates recurring cloud hosting expenditure, guarantees on-campus academic data sovereignty, and delivers sub-millisecond local LAN latency for lab workstations.

Cost-Free Campus Deployment

Deployment Invariant: OCI containerization guarantees identical microservice execution from local verification to Azure VM staging and final SIT campus bare metal production.

Future Scope: Autonomous IoT Lockers & Campus Integration

Architectural innovations from SDD Section 7 expanding digital governance to autonomous hardware and institutional university systems.

INNOVATION 01 — AUTONOMOUS CUSTODY

IoT Smart Lockers (RFID Relays)

ESP32 microcontrollers driving solenoid cabinet locks for 24/7 autonomous equipment checkout and return. Students swipe SIT RFID ID cards; system verifies active capstone loan approval and pops open the designated hardware bay without counter queues.

24/7 Hackathon & Lab Access

INNOVATION 02 — MOBILE SCANNER

Progressive Web App Barcode Scanner

High-speed camera vision scanning tool for lab assistants utilizing browser Barcode Detection APIs. Scans SIT serial barcodes (e.g. `SIT-ROBOT-0042`) directly on smartphones, accelerating physical counter handovers without optical hardware costs.

Zero Dedicated Scanner Costs

INNOVATION 03 — INSTITUTIONAL SYNC

Symbiosis Central SIS & LDAP Sync

Bi-directional integration with Symbiosis Central Student Information System and campus LDAP. Binds student PRNs to enrollment rosters, automates academic eligibility checks, and triggers automated graduation clearance holds for unreturned lab assets.

Automated Degree Clearance

Future Horizon: LabTrack evolves from a lab inventory system into an autonomous campus hardware operating system, bridging physical lockers with university enterprise networks.

Live System Demonstration & Verification Scorecard

INTERACTIVE CLIENT INTERFACE

Web Application • Port 5173

Full React 19 single-page application featuring student workstation, faculty approval workflow, counter dispatch barcode scanner, and defect triage bench.

LAUNCH LIVE UI → http://localhost:5173

BACKEND API DOCUMENTATION

FastAPI Interactive Swagger • Port 8000

Interactive OpenAPI endpoints demonstrating Bearer JWT authorization, atomic stock decrement transactions, outbox poller records, and schema validations.

OPEN SWAGGER DOCS → http://127.0.0.1:8000/docs

61 / 61	5 / 5	22 / 7	< 50ms
Automated Pytest Suites Passing	Docker Compose Containers Healthy	SQL Tables & Mongo Collections	ASGI Operational API Latency

Live Verification: The complete LabTrack platform is active, verified with 61 automated test suites, and ready for live examiner evaluation.

CLOUD COMPUTING ESSENTIALS (CCE) • SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

THANK YOU VERY MUCH!

OPEN FOR QUESTIONS & TECHNICAL VIVA DEFENSE

COMPREHENSIVE ACADEMIC LITERATURE REVIEW MATRIX

[1] AWS Prescriptive Guidance (2023): Transactional Outbox Pattern for Guaranteed Event Reliability across distributed microservices.

[2] Felter et al., IBM Research (IEEE 2015): An Updated Performance Comparison of Virtual Machines and Linux Containers (OCI isolation).

[3] Martin Fowler & Sadalage (2012): CQRS Pattern & NoSQL Distilled: Segregating ACID Relational Write Truth from BASE Document Read Feeds.

[4] Loper & Bird (2002) / TextBlob NLP: Lexical Semantics & Sentiment Analysis applied to Academic Requisition Justification Prioritization.

[5] IETF RFC 5321 (2008) & RFC 7519 (2015): SMTP Store-and-Forward Relay Architecture & Stateless JSON Web Token (JWT) Security Specifications.

[6] Snipe-IT & GLPI Systems (Comparative): Evaluation of open-source asset tools showing failure modes under university capstone concurrency.

COURSE EVALUATION • SIT PUNE

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Symbiosis Institute of Technology (SIT), Pune • Cloud Computing Essentials

CCE • 2 Credits • Official 30-Mark Rubric Evaluation